MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **WEEKLY CHAPTER TEST – 06 (LINEAR EQUATIONS)** | **Time:** 45 Minutes

Class: Class X (CBSE) | **Chapter 3: Pair of Linear Equations in Two Variables** | **MM:** 20

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choice is provided in Question 9(iv).
2. **Section A** comprises 3 Multiple Choice Questions (MCQs) of 1 mark each.
3. **Section B** comprises 1 Assertion-Reason question of 1 mark.
4. **Section C** comprises 2 Very Short Answer (VSAQ) questions of 2 marks each.
5. **Section D** comprises 1 Short Answer (3M) and 1 Long Answer / HOTS (5M) question.
6. **Section E** comprises 1 Applied Case Study with 4 sub-parts of 1 mark each.

SECTION A: Multiple Choice Questions (3 × 1 = 3 Marks)

Q1. The area of the triangle formed by the line $\frac{x}{a} + \frac{y}{b} = 1$ with the coordinate axes is: [1]

(A) ab (B) $2ab$ (C) $\frac{1}{2}ab$ (D) $\frac{1}{4}ab$

Q2. If $37x + 43y = 123$ and $43x + 37y = 117$, then the value of $x + y$ is: [1]

(A) 1 (B) 2 (C) 3 (D) 4

Q3. In the above problem ($37x + 43y = 123$ and $43x + 37y = 117$), the value of $x - y$ is: [1]

(A) 1 (B) -1 (C) 2 (D) -2

SECTION B: Assertion-Reason Question (1 × 1 = 1 Mark)

Q4. Assertion (A): If the system of equations $3x + 6y = 10$ and $2x - ky + 5 = 0$ is inconsistent, then $k = -4$.

Reason (R): The system $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ is inconsistent if $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false.
(D) (A) is false but (R) is true.

SECTION C: Very Short Answer Questions (2 × 2 = 4 Marks)

Q5. Solve the following pair of linear equations by substitution: [2]

$$s - t = 3 \quad \text{and} \quad \frac{s}{3} + \frac{t}{2} = 6$$

Q6. If the lines given by $3x + 2ky = 2$ and $2x + 5y + 1 = 0$ are parallel, determine the value of k. [2]

SECTION D: Short & Long Answer Questions $(1 \times 3 + 1 \times 5 = 8$ Marks)

Q7. Five years ago, Nuri was thrice as old as Sonu. Ten years later, Nuri will be twice as old as Sonu. How old are Nuri and Sonu? [3]

Q8. A chemist has one solution which is 50% acid and another solution which is 25% acid. How much of each should be mixed to make 10 litres of a 40% acid solution? [5]

SECTION E: Applied Case Study Based Question $(1 \times 4 = 4$ Marks)

Q9. Case Study: Eco-Tourism Boating Expedition & River Dynamics

An eco-tourism enterprise runs solar-powered safari boats on a river. The speed of the boat in still water is x km/h and the speed of the river current is y km/h. The downstream speed is $(x + y)$ km/h and the upstream speed is $(x - y)$ km/h. During a test trial, the boat covers 21 km downstream in 3 hours, and on the return trip upstream, it covers 12 km in 4 hours.

Based on the above information, answer the following questions:

(i) Formulate the pair of linear equations in x and y representing the downstream and upstream journeys. [1]

(ii) Find the speed of the safari boat in still water (x). [1]

(iii) Find the speed of the river current (y). [1]

(iv) What total time will the boat take to travel 10 km downstream and then return back 10 km upstream? [1]

OR

Find the ratio of downstream speed to upstream speed. [1]

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★★★ END OF QUESTION PAPER · ALL THE BEST ★★★